
Research Scientist at NVIDIA Research, working on efficient and scalable AI, with a recent focus on recursive self-improvement (RSI) for coding agents and large-scale experimentation across thousands of coding environments and agent instances. Hands-on R&D experience in large-scale multimodal generation across image and video. Previously worked with Meta GenAI / BizAI, contributing to the successful delivery of ImageGen and VideoGen products and large-scale foundation-model systems. Experienced in 1K+ GPU-scale pretraining, text-to-image generation, image-to-video generation, autoregressive diffusion, and efficient generative modeling. First-authored 10+ top-tier publications with 1,000+ citations and received a Workshop Best Paper Award. Received a Ph.D. in Computer Science from KAUST, advised by Prof. Juergen Schmidhuber.

EDUCATION

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- **KAUST – AI Initiative** Saudi Arabia
Ph.D. in CS (Advisor: Juergen Schmidhuber) Aug. 2022 – Mar. 2026
 - GPA: 4.0/4.0
 - Research Interests: Explainable, Efficient, and Responsible Generative Models

SELECTED PUBLICATIONS

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1. **Liu, H.**, Liu, D., Zhuge, M., Zhou, Z., Xie, T., He, S., et al. & Schmidhuber, J. Mixture of States: Routing Token-Level Dynamics for Multimodal Generation. CVPR 2026. [\[pdf\]](#)
 2. **Liu, H.**, Liu, S., Zhou, Z., et al. & Perez-Rua, J. M. MarDini: Masked Autoregressive Diffusion for Video Generation at Scale. TMLR. [\[pdf\]](#)
 3. **Liu, H.**, Zhang, W., Xie, J., Faccio, F., Xu, M., Xiang, T., Shou, M., Perez-Rua, J. M. & Schmidhuber, J. Faster Diffusion Through Temporal Attention Decomposition. ICLR 2026 & TMLR. [\[pdf\]](#) [\[code\]](#)
 4. **Liu, H.**, Zhang, W., Li, B., Ghanem, B. & Schmidhuber, J. “Lazy” Layers Make Fine-tuned Diffusion Models More Traceable. Technical Report. [\[pdf\]](#)
 5. **Liu, H.**, Zhuge, M., Li, B., Wang, Y., Faccio, F., Ghanem, B. & Schmidhuber, J. Learning to Identify Critical States for Reinforcement Learning from Videos. ICCV 2023. [\[pdf\]](#) [\[code\]](#)
 6. **Liu, H.**, Zhang, W., Li, B., Wu, H., He, N., Huang, Y., Li, Y., Ghanem, B. & Zheng, Y. AdaptiveMix: Improving GAN Training via Feature Space Shrinkage. CVPR 2023. [\[pdf\]](#) [\[code\]](#)
 7. **Liu, H.**, Li, B., Wu, H., Liang, H., Huang, Y., Li, Y., et al. & Zheng, Y. Combating Mode Collapse in GANs via Manifold Entropy Estimation. AAAI 2023. [\[pdf\]](#) [\[code\]](#)
 8. **Liu, H.**, Wu, H., Xie, W., Liu, F. & Shen, L. Group-wise Inhibition-based Feature Regularization for Robust Classification. ICCV 2021. [\[pdf\]](#) [\[code\]](#)
 9. **Liu, H.**, Zhang, W., Liu, F., Wu, H. & Shen, L. Fingerprint Presentation Attack Detector Using Global-Local Model. IEEE Transactions on Cybernetics. [\[pdf\]](#) [\[code\]](#)
 10. **Liu, H.**, Zhang, W., Xie, J., Wu, H., Li, B., Zhang, Z., Li, Y., Huang, Y., Ghanem, B. & Zheng, Y. Decoupled Mixup for Out-of-Distribution Visual Recognition. ECCV 2022 Workshop. [\[pdf\]](#) [\[code\]](#)
 11. Zhuge, M., Zhao, C., **Liu, H.**, Zhou, Z., Liu, S., Wang, W., Chang, E., Le Lan, G., Fei, J., Zhang, W., et al. & Schmidhuber, J. Neural Computers. Technical Report. **Main Contributor.** [\[pdf\]](#) [\[code\]](#)
 12. Zhu, H., **Liu, H.**, Zhao, Y., Ye, T., Chen, J., et al. SANA-WM: Efficient Minute-Scale World Modeling with Hybrid Linear Diffusion Transformer. arXiv:2605.15178, 2026. **Equal Contribution.** [\[pdf\]](#) [\[project\]](#)
 13. Zhang, W., **Liu, H.**, Li, B., et al., Zheng, Y. & Ghanem, B. Dynamically Masked Discriminator for Generative Adversarial Networks. NeurIPS 2023. **Equal Contribution.** [\[pdf\]](#) [\[code\]](#)
 14. Zhang, W., **Liu, H.**, Liu, F., Ramachandra, R. & Busch, C. Effective Presentation Attack Detection Driven by Face Related Task. ECCV 2022. **Equal Contribution.** [\[pdf\]](#) [\[code\]](#)
 15. Zhuge, M., **Liu, H.**, et al. & Schmidhuber, J. Mindstorms in Natural Language-Based Societies of Mind. NeurIPS Workshop 2023. **Best Workshop Paper, Equal Contribution.** [\[pdf\]](#) [\[code\]](#)

AWARDS, GRANTS & HONORS

Workshop Best Paper Award	2023
Co-PI, “Efficient Generative Models”, SDAIA–KAUST Grant (200k SAR)	2023

RESEARCH EXPERIENCE

- **Efficient AI, NVIDIA Research** Singapore
Research Scientist working with Prof. Song Han 2026 – present
 - Research Topics: Recursive self-improvement, coding agents, efficient video generation, long-horizon world models, and full-stack inference acceleration.
 - **Research Mentorship:** Co-mentored Tian Ye and Haoyi Zhu with Dr. Enze Xie.
 - Equal-contribution author of **SANA-WM**, a 2.6B open-source world model for controllable, minute-long 720p video generation on a single GPU.
 - Co-authored **SANA-Video 2.0** and **SANA-Streaming**, enabling efficient 720p generation and real-time 1280×704 video editing at 24 FPS on a single RTX 5090.
 - Co-authored **Sol-Engine** and **Sol-Attn**, training-free full-stack acceleration for video diffusion with $2.27\text{--}2.77\times$ end-to-end speedups on benchmarked models.
- **BizAI, Meta** MPK, United States
Research Scientist Intern working with Ding Liu and Wei Liu 2025
 - Research Topic: Large-scale pretraining, text-to-image generation, mixed-modality generation.
 - Publication Records: CVPR $\times 4$.
 - Co-developed a commercial 7B text-to-image model trained on over 1B image–text pairs and deployed in several widely used products.
 - Built a 20B native LLM-driven multimodal generation prototype from scratch.
 - Contributed to a native unified model for both visual understanding and generation tasks.
 - Contributed to Neural Computer Project.
- **MGenAI, Meta** London, United Kingdom
Research Scientist Intern working with J.-M. Perez-Rua and T. Xiang 2024
 - Research Topic: Foundational training, image-to-video generation, text-to-image generation.
 - Publication Records: TMLR $\times 2$; ICCV $\times 1$; CVPR $\times 1$.
 - Co-developed a foundational text-to-image model to support several well-known products.
 - Scaled autoregressive diffusion to video generation.
- **GenAI, KAUST** Saudi Arabia
Ph.D. Candidate supervised by Prof. Juergen Schmidhuber 2022 – 2026
 - Research Topic: Neural networks with multiple-step inference, including diffusion models, autoregressive models, and RL agents.
 - Publication Records: ICCV $\times 1$; CVPR $\times 2$; NeurIPS Workshop $\times 1$; TMLR $\times 2$; under review $\times 1$.
 - Highlight: NLSOM received a **Workshop Best Paper Award**.
 - Highlight: T-GATE was merged into the Diffusers library and received over **300** stars on GitHub.
- **Jarvis Lab, Tencent** Shenzhen, China
Intern working with N. He, Y. Li, and Y. Zheng 2021 – 2022
 - Research Topic: Generative models and medical imaging.
 - Publication Records: NeurIPS $\times 1$; CVPR $\times 1$; ICCV $\times 1$; AAAI $\times 1$; MICCAI $\times 2$; MIA $\times 1$; PR $\times 1$; ECCVW $\times 1$.
 - Highlight: MaF-GAN was recognized as an **Oral** paper at AAAI.
 - Highlight: Ranked 4th in the ECCV 2022 NICO Challenge.